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14、

(a) $\sin^2\theta\cos^2\varphi = \frac{1}{2}[1+\cos 2\varphi]\sin^2\theta = \frac{1}{2}\left[S_{00}-S_{20}+\frac{1}{3}S_{22}(\theta,\varphi)\right]$

(b) $= (1+\cos\theta)\sin\theta\cos\varphi = -S_{11}(\theta,\varphi)-\frac{1}{3}S_{21}(\theta,\varphi)$

15、

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{du}{dr}\right)+\frac{1}{r^2\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{du}{d\theta}\right)+\frac{1}{r^2\sin^2\theta}\frac{d^2u}{d\varphi^2}=A+Br^2\sin2\theta\cos\varphi$$

$u|_{\phi=0}=u|_{\phi=2\pi}$, $\frac{du}{d\varphi}\Big|_{\varphi=0}=\frac{du}{dr}\Big|_{\varphi=2\pi}$

$u|_{\theta=0}$有界 $u|_{\theta=\pi}$有界

$u|_{r=0}$有界 $u|_{r=a}=0$

将u, f根据本征函数展开

$$u(r,\theta,\varphi)=X(r)S_{00}(\theta,\varphi)+Y(r)S_{21}(\theta,\varphi)$$

$$f(r,\theta,\varphi)=\sum_{n=0}^{\infty}g_n(r)S_{ln}(\theta,\varphi)=AS_{00}(\theta,\varphi)-\frac{2}{3}Br^2S_{21}(\theta,\varphi)$$

代入方程边界得

$$\begin{cases}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dX(r)}{dr}\right)=A\\ \frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dY(r)}{dr}\right)-\frac{Y(r)}{r^2}\cdot6=-\frac{2}{3}Br^2\end{cases}$$

且 $\begin{cases}X(0),\ Y(0)\text{有界}\\ X(a)=Y(a)=0\end{cases}$

得: $X=\frac{1}{6}A(r^2-a^2)$, $Y=\frac{1}{21}B(r^2-a^2)r^2$

$\Rightarrow u(r,\theta,\varphi)=\frac{1}{6}A(r^2-a^2)+\frac{1}{14}B(r^2-a^2)r^2\sin2\theta\cos\varphi$

A.7

(b) 代入逐项积

$$\int_0^\infty e^{-ax} J_0(\sqrt{bx})dx = \int_0^\infty e^{-ax} \sum_{k=0}^\infty \frac{(-1)^k}{(k!)^2}\left(\frac{bx}{4}\right)^k dx$$

$$= \sum_{k=0}^\infty \frac{(-1)^k}{(k!)^2}\left(\frac{b}{4}\right)^k \int_0^\infty e^{-ax} x^k dx$$

$$= \sum_{k=0}^\infty \frac{(-1)^k}{k!}\left(\frac{b}{4}\right)^k \frac{1}{a^{k+1}}$$

$$= \frac{1}{a}\sum_{k=0}^\infty \left(\frac{b}{a}\right)^k \left(\frac{-1}{4}\right)^k \cdot \frac{1}{k!}$$

$$= \frac{1}{a} e^{-\frac{b}{4a}}$$

(c) $$\int_0^\infty e^{-ax} J_\nu(bx) x^{\nu+1} dx = \int_0^\infty e^{-ax} \sum_{k=0}^\infty \frac{(-1)^k}{k!\,\Gamma(k+\nu+1)}\left(\frac{bx}{2}\right)^{2k+\nu} x^{\nu+1} dx$$

$$= \sum_{k=0}^\infty \frac{(-1)^k}{k!\,\Gamma(k+\nu+1)}\left(\frac{b}{2}\right)^{2k+\nu} \int^\infty e^{-ax} x^{2k+2\nu+1} dx \quad \frac{1}{a^{2k+2\nu+2}}$$

$$= \frac{1}{a^2}\left(\frac{b}{2a^2}\right)^\nu \sum_{k=0}^\infty \frac{\Gamma(2k+2\nu+2)}{k!\,\Gamma(k+\nu+1)}\left(\frac{-b^2}{4a^2}\right)^k$$

$$= \frac{2}{\sqrt{\pi}\,a^2}\left(\frac{4b}{a^2}\right)^\nu \sum_{k=0}^\infty \frac{1}{k!}\Gamma\left(k+\nu+\frac{3}{2}\right)\left(\frac{-b^2}{a^2}\right)^k$$

$$= \frac{2}{\sqrt{\pi}\,a^2}\left(\frac{4b}{a^2}\right)^\nu \sum_{k=0}^\infty \frac{\left(k+\nu+\frac{1}{2}\right)\cdots\left(\nu+\frac{3}{2}\right)\Gamma\left(\nu+\frac{3}{2}\right)}{k!}\left(\frac{-b^2}{a^2}\right)^k$$

$$= \frac{2}{\sqrt{\pi}\,a^2}\left(\frac{4b}{a^2}\right)^\nu \left(1-\frac{b^2}{a^2}\right)^{-\frac{3}{2}} \Gamma\left(\nu+\frac{3}{2}\right)$$